Practical 07

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Repo link - https://github.com/IT24610788/IT3012---Practical-Base.git

1. Reachability vs. Feasibility

Reachability evaluates physical map constraints (e.g., boundaries, wall collisions) determined by the environment geometry. Feasibility evaluates logical constraints (e.g., safety, threat level) determined by the rules of the game. The KB enforces feasibility by feeding tile-specific percepts into the inference engine; if the rule engine deduces a prohibitive state (like 'Retreat'), the A* algorithm discards the otherwise physically reachable node.

2. Declarative vs. Procedural Paradigms

Procedural nested if-else statements hardcode logic into the control flow, creating brittle, monolithic "spaghetti code" that becomes unmaintainable at scale. The declarative KB approach isolates the *knowledge* (the rules) from the *reasoning algorithm* (forward chaining). If 50 new game rules are added, the agent's movement code remains entirely untouched; only new data structures are registered.

3. Modus Ponens & Horn Clauses

Modus Ponens dictates that if $P \implies Q$, and P is true, then Q is true. The forward_chain loop acts as this mathematical execution via the all(premise in self.facts for premise in premises) check. If all antecedents (P) evaluate to true within the known facts set, the engine fires the rule and appends the consequent (Q) to the facts set.